



OPERATING SYSTEM DESIGN

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Department of Computer Applications

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Memory Management and File System Management

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2^1 Bytes = 2 Bytes

2^2 Bytes = 4 Bytes

2^3 Bytes = 8 Bytes

2^4 Bytes = 16 Bytes

2^5 Bytes = 32 Bytes

2^6 Bytes = 64 Bytes

2^7 Bytes = 128 Bytes

2^8 Bytes = 256 Bytes

2^9 Bytes = 512 Bytes

2^{10} Bytes = 1024 Bytes = 1KB

2^{20} Bytes = 1024 KB = 1 MB

2^{30} Bytes = 1024 MB = 1 GB

Assuming a 1 KB page size, what are the page numbers (starts at 0) and offsets for the following address references (provided as decimal numbers):

a. 2375

b. 19366

c. 30000

d. 256

e. 16385

Calculate the size of memory if its address consists of 22 bits and the memory is 2-byte addressable

Calculate the size of memory if its address consists of 22 bits and the memory is 2-byte addressable

If the memory address is n bits then, 2^n * size of one location

$$2^{22} * 2 \text{ byte} = 2^{23} \text{ bytes} = 8 \text{ MB}$$

Calculate the number of bits required in the address for memory having size of 8 GB. Assume the memory is 4-byte addressable.

Calculate the number of bits required in the address for memory having size of 8 GB. Assume the memory is 4-byte addressable

$$2^n * 4 \text{ byte} = 8 \text{ GB}$$

$$2^n * 2^2 \text{ bytes} = 2^{33}$$

$$2^n = 2^{31}$$

$$n = 31 \text{ bits}$$

Given, LAS =128 KB, PAS=512 KB, Page size=16KB, 1 byte Addressble

How many Number of bits required for LA?

How many Number of bits required for PA?

How many pages are there in the program?

What is the frame size?

How many entries are there in the Pagetable?

What is the size of the pagetable?

How does the LA be splitted?

How does the PA be splitted?

Number of bits required for LA

$$LAS = 128 \text{ KB} = 2^{17}$$

$$N=17 \text{ bits} \Rightarrow LA = 17 \text{ bits}$$

Number of bits required for PA

$$PAS = 512 \text{ KB} = 2^{19}$$

$$N=19 \text{ bits PA}=19 \text{ bits}$$

$$\text{Page size}=16\text{KB} = 2^{14}$$

How many pages are there in the program?

$$LAS/\text{page size} = 2^{17}/2^{14} = 2^3 = 8 \text{ pages}$$

What is the frame size?

$$\text{Page size} = \text{frame size} = 2^{14} \text{ KB}$$

How many entries are there in the Pagetable?

$$8 \text{ pages} = 8 \text{ entries}$$

What is the size of the pagetable?

$$8 \text{ pages of Each } 1 \text{ byte addressable, } 8 \text{ bytes}$$

How is LA splitted ?

Logical Address = p, d

LA = 17 bits $\Rightarrow p + d = 17$ bits

No of Pages = 8 = $2^3 \Rightarrow 3$ bits

$3 + 14 = 17$ bits $\Rightarrow p = 3$ bits, $d = 14$ bits

How is PA splitted ?

Logical Address = f, d

PA = 19 bits $\Rightarrow f + d = 19$ bits

$d = 14$ bits $\Rightarrow f = 5$ bits

How is LA splitted ?

Logical Address = p, d

LA = 17 bits $\Rightarrow p + d = 17$ bits

No of Pages = 8 = $2^3 \Rightarrow 3$ bits

$3 + 14 = 17$ bits $\Rightarrow p = 3$ bits, $d = 14$ bits

How is PA splitted ?

Logical Address = f, d

PA = 19 bits $\Rightarrow f + d = 19$ bits

$d = 14$ bits $\Rightarrow f = 5$ bits

Given, $LAS = 64 \text{ KB}$, $PAS = 128 \text{ KB}$, Page size = 4 KB , 1 byte Addressable

How many Number of bits required for LA?

How many Number of bits required for PA?

How many pages are there in the program?

What is the frame size?

How many entries are there in the Pagetable?

What is the size of the Pagetable?

How does the LA be splitted?

How does the PA be splitted?

Given, LAS = 512 KB,

No of Pages = 1 KB

Number of frames = 256 Byte

Page offset = 10

Memory is 1 byte addressable

Calculate PAS, Page Size and Page Table Size

Consider a system has 4K pages of 512 bytes in size in logical address space. How many bits are required for logical address?

Consider a logical address space of 64 pages of 1024 words each, mapped on to a physical memory of 32 frames.

How many bits are there in the physical address and logical address?



THANK YOU

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